EE652

IC Design Laboratory

RISCV Phase 2 Submission

Simulation Results

Group 3

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Work Distribution

R-Type: Kshitij Agarwal, Krish Patel
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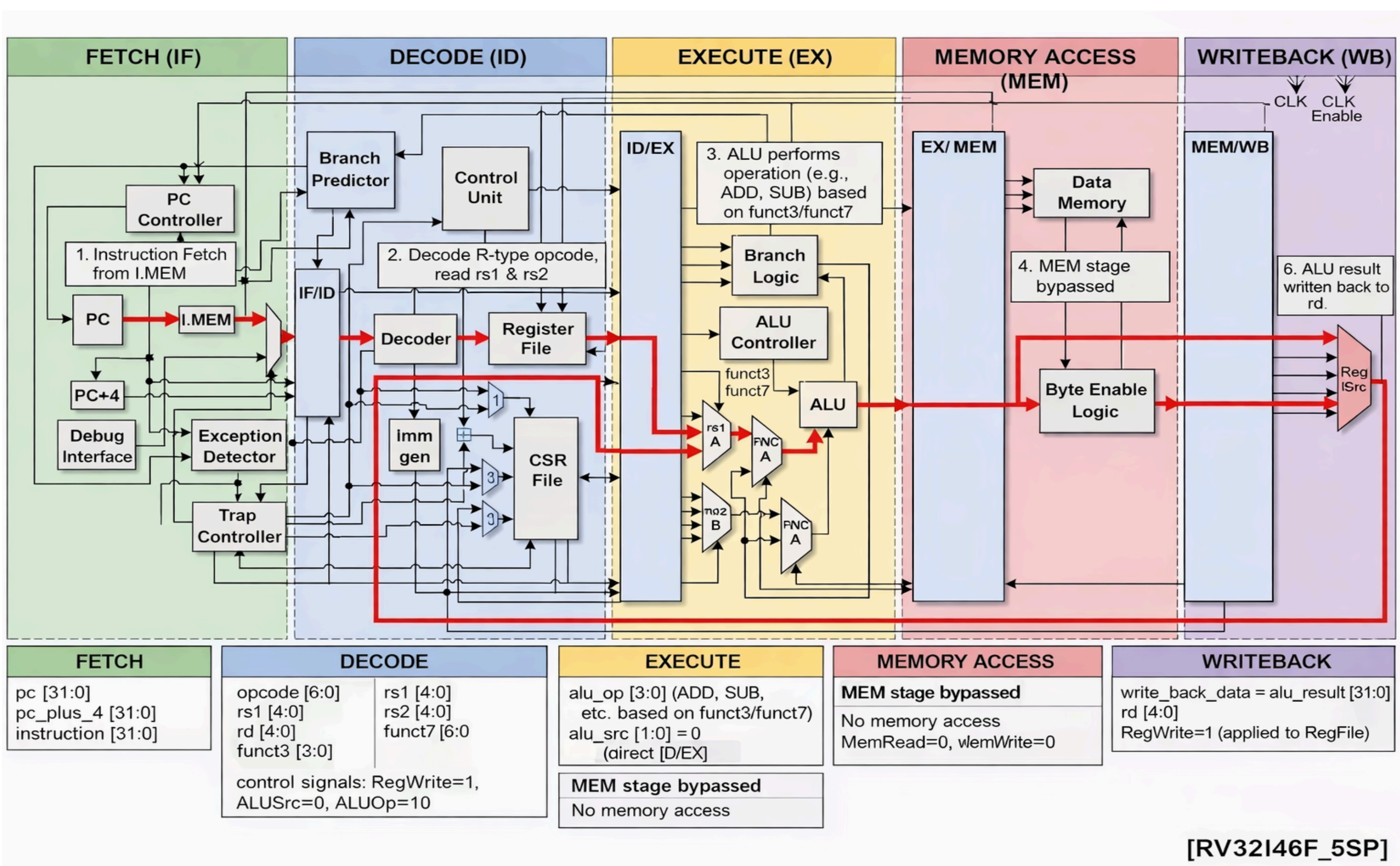
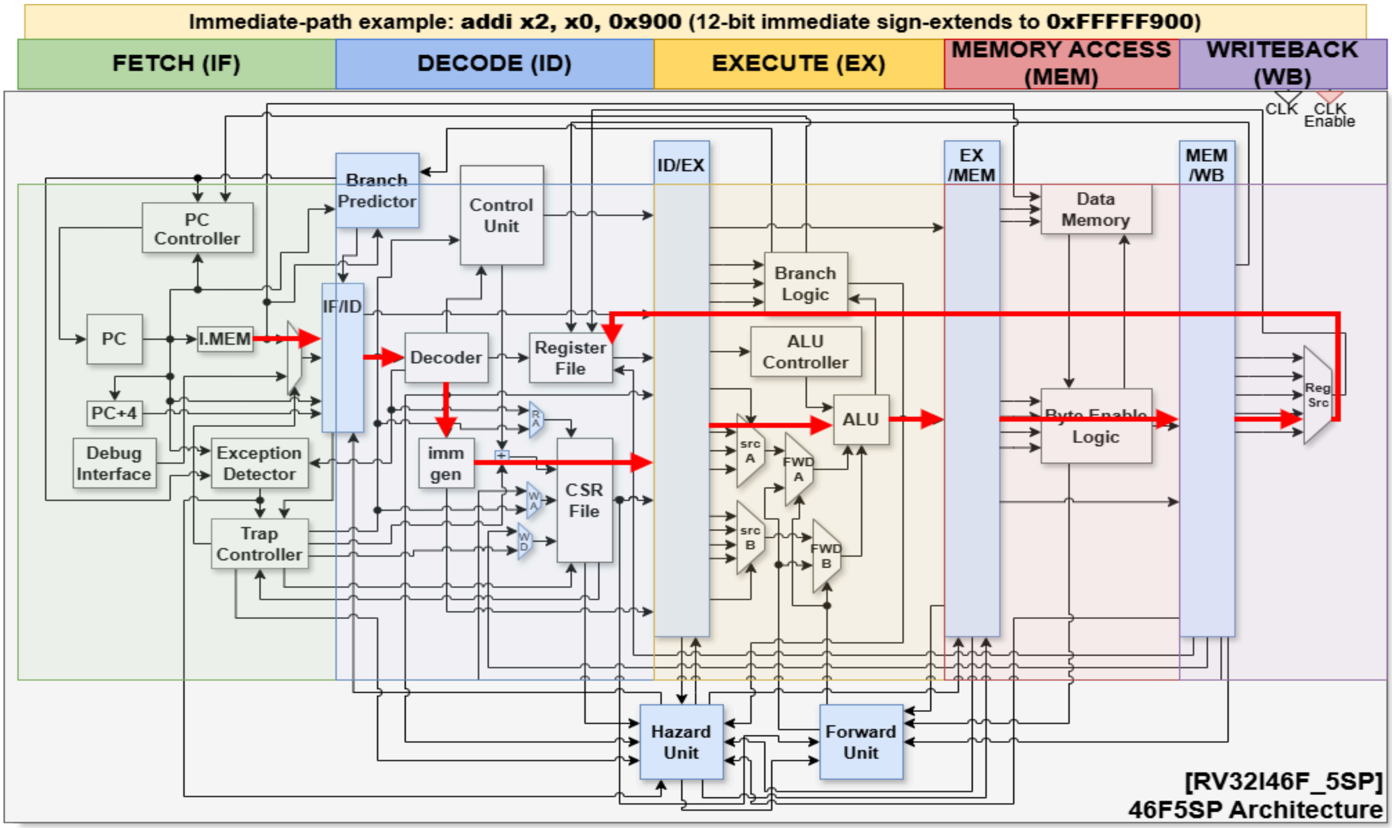
B-Type, J-type: Shail Joshi, Shriniket Behera
S-Type, U-Type: Anirudh Mittal, Tirth Shah
Hazards: Arjun Anand Mallya, Rishank Soni

Block Diagram: R- Type



Block Diagram: I- Type



5-stage / 5-clock movement of the same instruction

Clock 1 IF	Clock 2 ID	Clock 3 EX	Clock 4 MEM	Clock 5 WB
FETCH PC selects instruction. Instruction enters IF/ID register.	DECODE I-type opcode, rs1=x0, rd=x2, imm[11:0]=0x900 -> 0xFFFFF900. RegWrite=1, ALUSrcB=IMM	EXECUTE ALU performs ADD. 0 + 0xFFFFF900 = 0xFFFFF900	MEMORY ACCESS No memory access. Result only passes through EX/MEM and MEM/WB.	WRITEBACK WB_alu_result is selected. Register file writes x2 = 0xFFFFF900

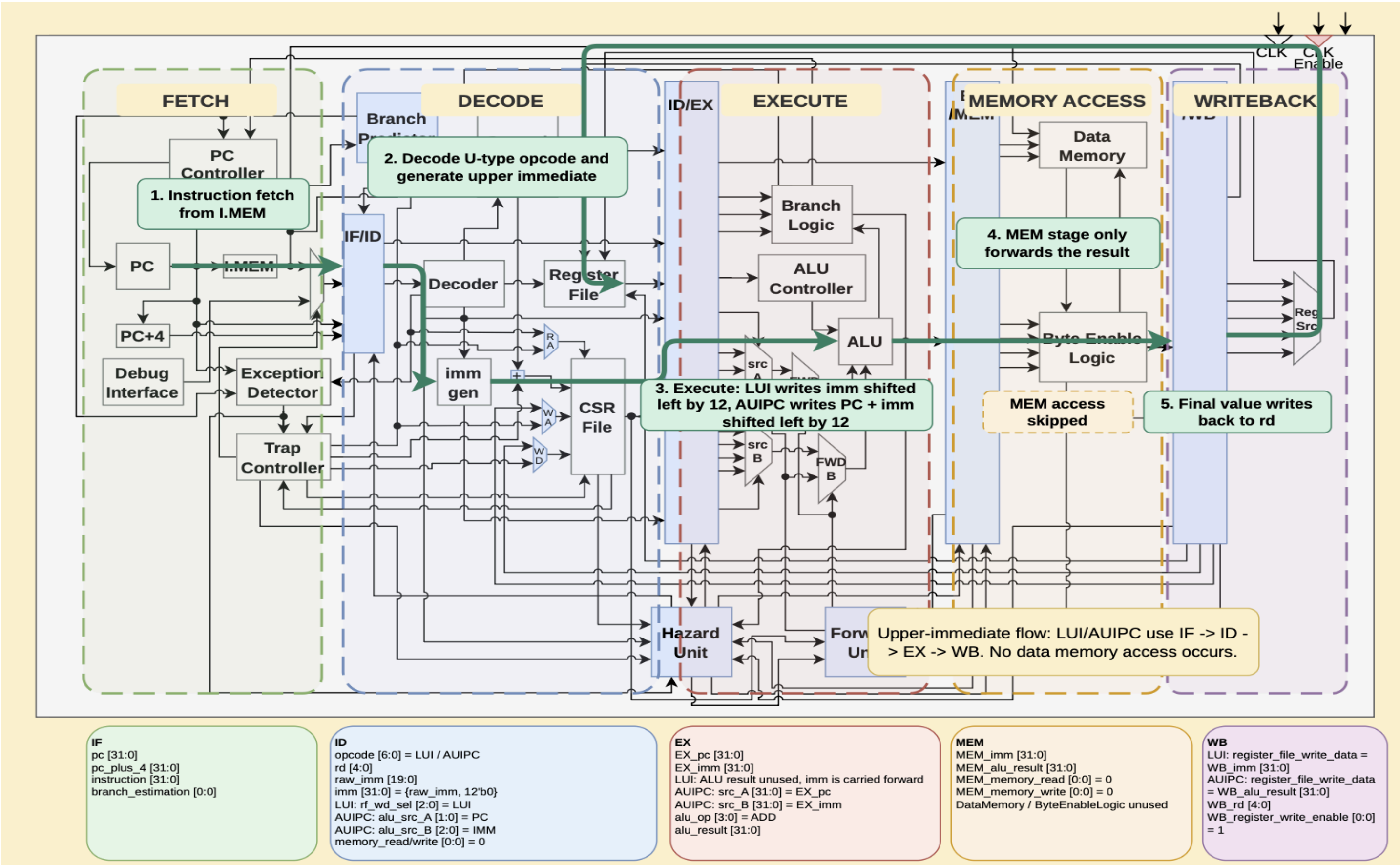
Important signal names and bit widths

IF	ID	EX	MEM	WB
pc[31:0] pc_plus_4_signal[31:0] instruction[31:0]	opcode[6:0], funct3[2:0] rs1[4:0], rd[4:0] raw_imm[19:0], imm[31:0] read_data1[31:0]	EX_imm[31:0] src_A[31:0], src_B[31:0] alu_op[3:0] alu_result[31:0]	MEM_alu_result[31:0] MEM_memory_read MEM_memory_write write_mask[3:0]	WB_alu_result[31:0] WB_rd[4:0] WB_register_write_enable register_file_write_data[31:0]

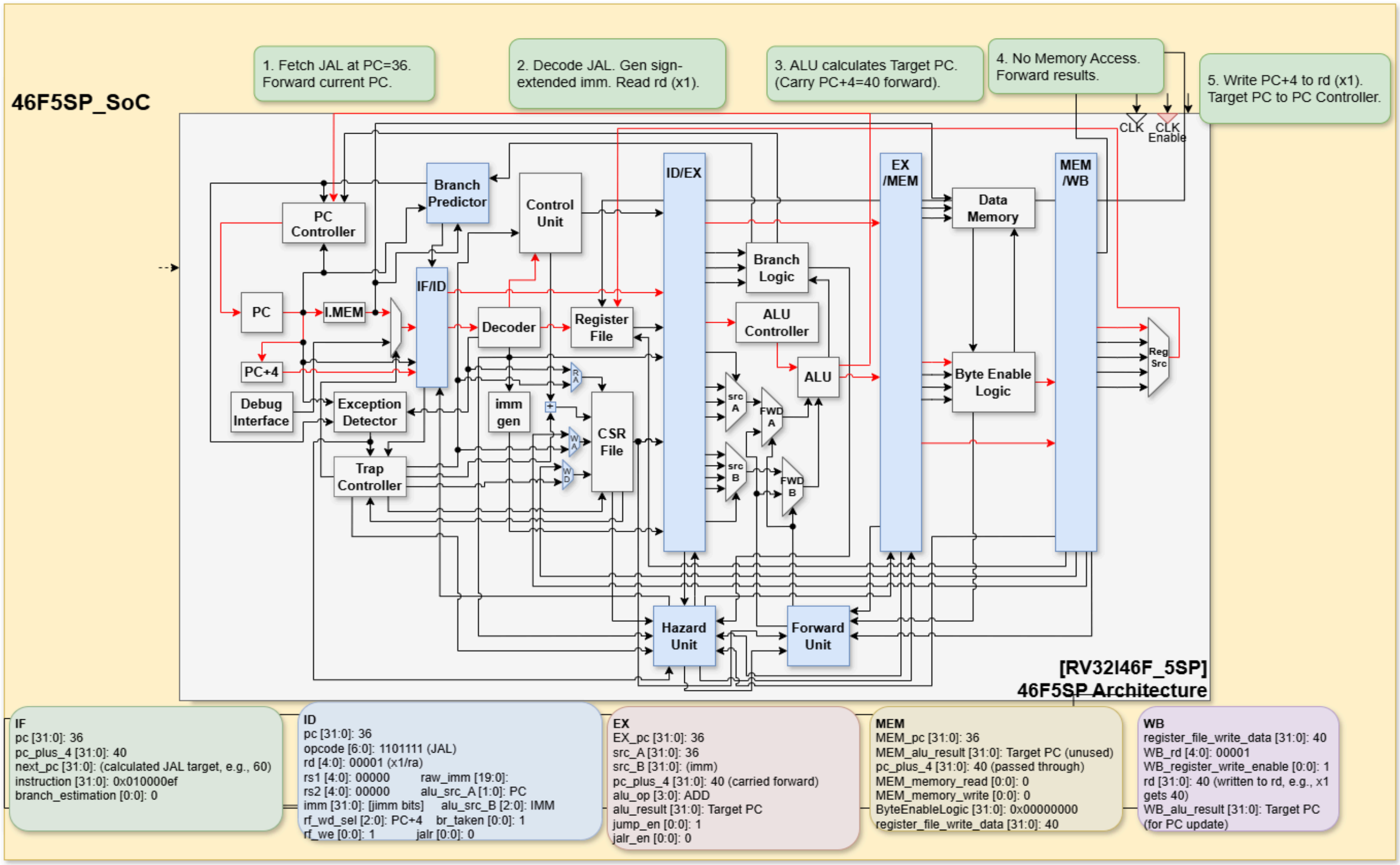
Pipeline registers
IF/ID: ID_pc[31:0], ID_instruction[31:0]
ID/EX: EX_rd[4:0], EX_imm[31:0], EX_read_data1[31:0]
EX/MEM: MEM_alu_result[31:0], MEM_rd[4:0]
MEM/WB: WB_alu_result[31:0], WB_rd[4:0]

addi control: alu_src_B_select=IMM, register_file_write=1, memory_read=0, memory_write=0

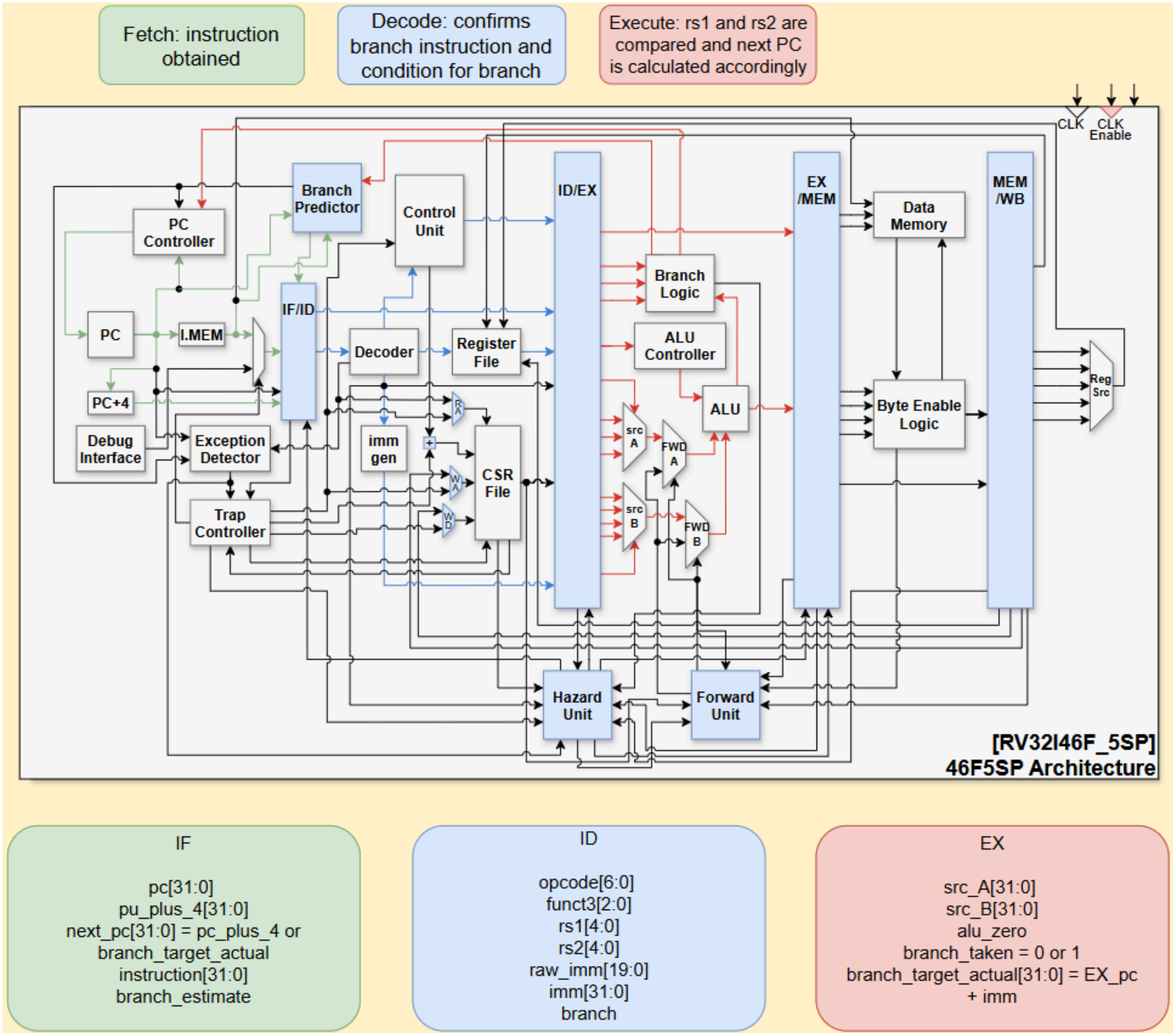
Block Diagram: U- Type


CLK
Clk
Enable
FETCH
DECODE
ID/EX
EXECUTE
MEMORY ACCESS
WRITEBACK
Branch
PC
Controller
1. Instruction fetch
from I.MEM
2. Decode U-type opcode and
generate upper immediate
IF/ID
PC
I.MEM
PC+4
Debug
Interface
Exception
Detector
Trap
Controller
Decoder
Register
File
imm
gen
CSR
File
Branch
Logic
ALU
Controller
ALU
src
A
src
B
FWD
B
3. Execute: LUI writes imm shifted
left by 12, AUIPC writes PC + imm
shifted left by 12
Data
Memory
4. MEM stage only
forwards the result
Byte Enable
Logic
MEM access
skipped
5. Final value writes
back to rd
Reg
Src
Hazard
Unit
Upper-immediate flow: LUI/AUIPC use IF -> ID -
> EX -> WB. No data memory access occurs.
IF
pc [31:0]
pc_plus_4 [31:0]
instruction [31:0]
branch_estimation [0:0]
ID
opcode [6:0] = LUI / AUIPC
rd [4:0]
raw_imm [19:0]
imm [31:0] = {raw_imm, 12'b0}
LUI: rf_wd_sel [2:0] = LUI
AUIPC: alu_src_A [1:0] = PC
AUIPC: alu_src_B [2:0] = IMM
memory_read/write [0:0] = 0
EX
EX_pc [31:0]
EX_imm [31:0]
LUI: ALU result unused, imm is carried forward
AUIPC: src_A [31:0] = EX_pc
AUIPC: src_B [31:0] = EX_imm
alu_op [3:0] = ADD
alu_result [31:0]
MEM
MEM_imm [31:0]
MEM_alu_result [31:0]
MEM_memory_read [0:0] = 0
MEM_memory_write [0:0] = 0
DataMemory / ByteEnableLogic unused
WB
LUI: register_file_write_data =
WB_imm [31:0]
AUIPC: register_file_write_data
= WB_alu_result [31:0]
WB_rd [4:0]
WB_register_write_enable [0:0]
= 1

<u>Block Diagram: J- Type</u>

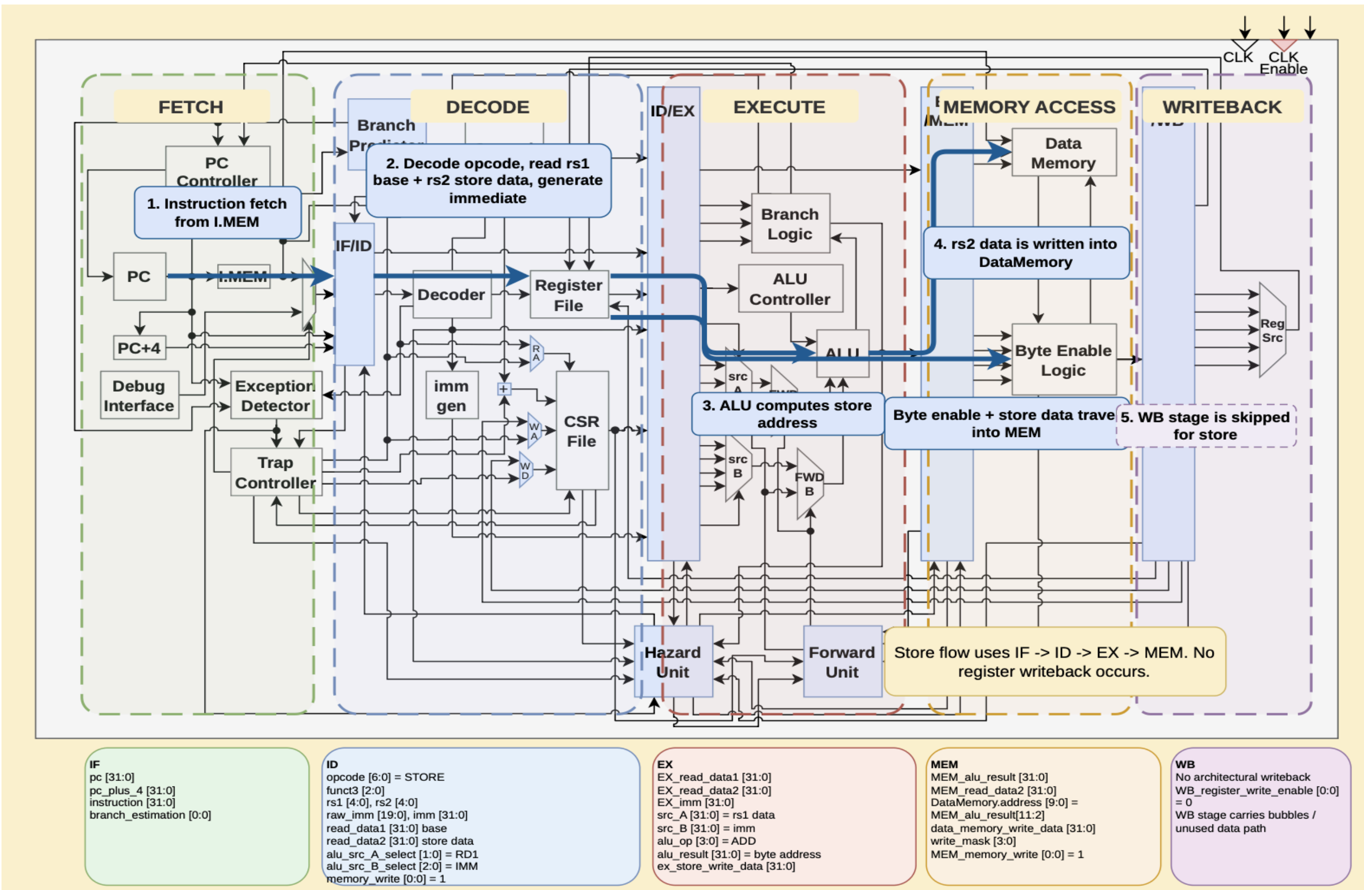


<u>Block Diagram: B- Type</u>

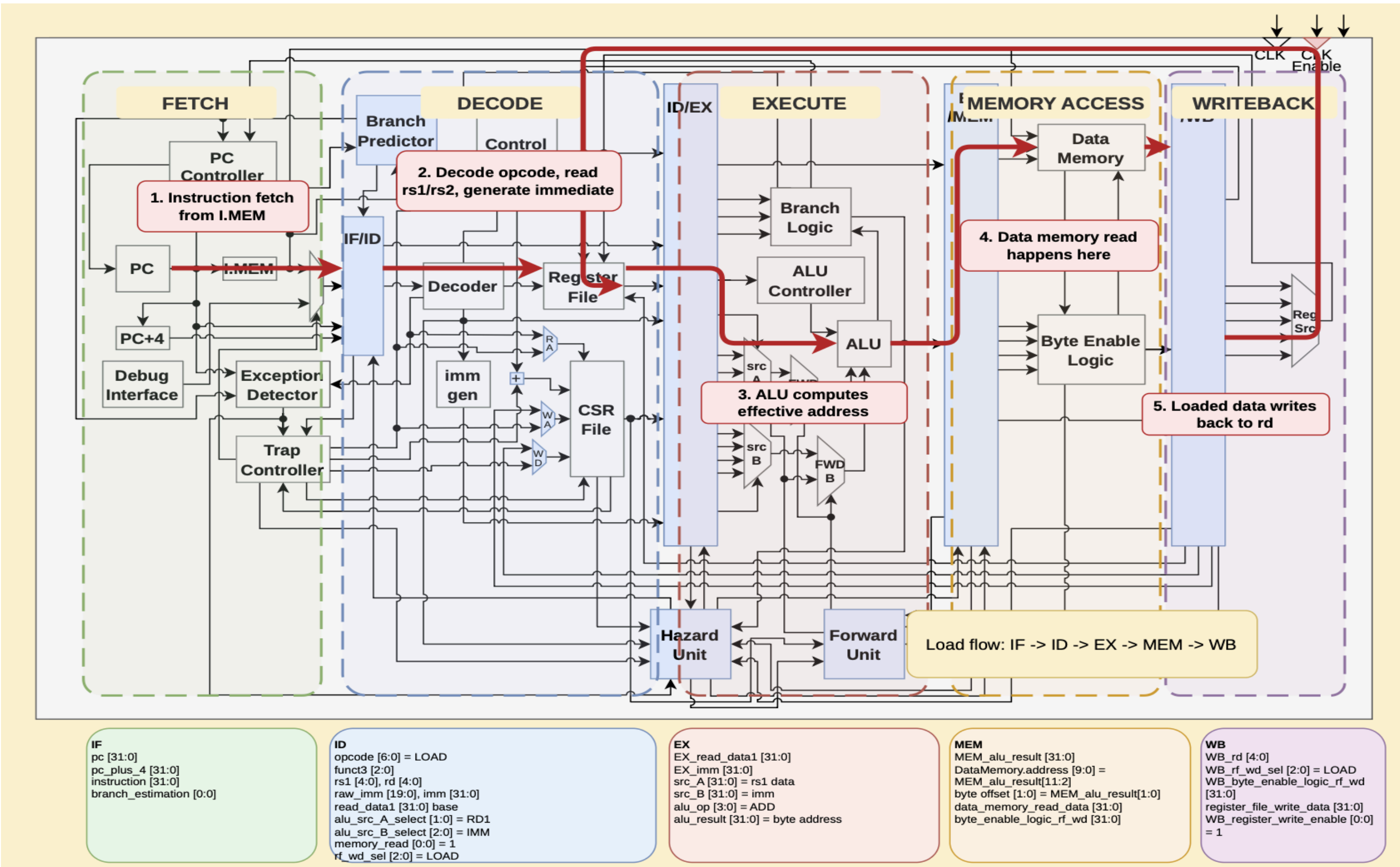


Fetch: instruction obtained
Decode: confirms branch instruction and condition for branch
Execute: rs1 and rs2 are compared and next PC is calculated accordingly
CLK
CLK Enable
Branch Predictor
Control Unit
ID/EX
EX /MEM
MEM /WB
Data Memory
PC Controller
Branch Logic
IF/ID
PC
I.MEM
Decoder
Register File
ALU Controller
ALU
PC+4
Byte Enable Logic
Reg Src
Debug Interface
Exception Detector
imm gen
CSR File
src A
FWD A
src B
FWD B
Trap Controller
Hazard Unit
Forward Unit
[RV32I46F_5SP]
46F5SP Architecture
IF
pc[31:0]
pu_plus_4[31:0]
next_pc[31:0] = pc_plus_4 or branch_target_actual
instruction[31:0]
branch_estimate
ID
opcode[6:0]
funct3[2:0]
rs1[4:0]
rs2[4:0]
raw_imm[19:0]
imm[31:0]
branch
EX
src_A[31:0]
src_B[31:0]
alu_zero
branch_taken = 0 or 1
branch_target_actual[31:0] = EX_pc + imm

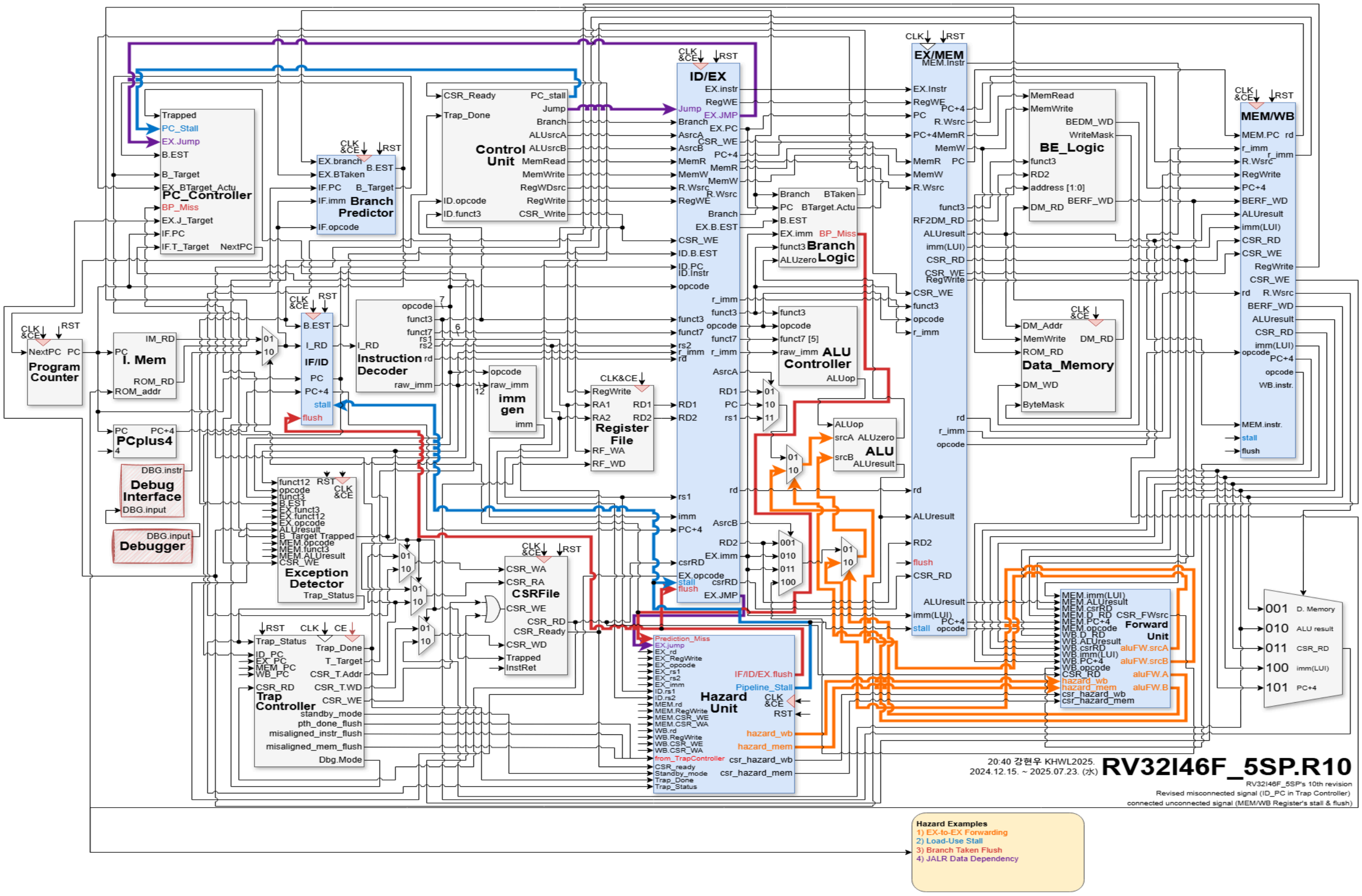
Block Diagram: S- Type



Block Diagram: Load Instruction



Block Diagram: Hazards



Ports Table

Module: RV32I46F5SP

Port Name	Width / Type	Source Module	Explanation
clk	1-bit Input	External System	he master clock signal that synchronizes all pipeline stage
reset	1-bit Input	External System	ctive-high reset signal used to initialize the processor stat

Port Name	Width / Type	Destination Module	Explanation
retire_instruction	32-bit Output	External Debugger	tion currently completing execution (writing back verification.

1. **Module: ProgramCounter**

Signal Table

Port Name	Width / Type	Source Module	Explanation
clk	1-bit Input	System Clock	The clock edge trigger for updating the PC value.
reset	1-bit Input	System Reset	Resets the PC to the boot address (0x0).

next_pc	32-bit Input	PC Controller	The calculated address of the next instruction to fetch.

Port Name	Width / Type	Destination Module	Explanation
pc	32-bit Output	Instruction Memory / PC+4	e current address of the instruction being execute

1. **Module: PCPlus4**

Signal Table

Port Name	Width / Type	Source Module	Explanation
pc	32-bit Input	Program Counter	The current Program Counter value.

Port Name	Width / Type	Destination Module	Explanation
pc_plus_4	32-bit Output	F/ID Register / PC Controlle	address of the next sequential instruction (pc +

1. **IF_ID_Register Module**

Signal Table

Port Name	Width / Type	Source Module	Explanation

clk	1-bit Input	System Clock	ck signal for synchronising the pipeline sta
reset	1-bit Input	System Reset	esets the register contents to default (NOP)
flush	1-bit Input	Hazard Unit	ears the register (inserts a bubble) when hig
IF_ID_stall	1-bit Input	Hazard Unit	lds current value (freezes pipeline) when hi
IF_pc	32-bit Input	Program Counter	PC value from the Fetch stage.
IF_pc_plus_4	32-bit Input	PC Plus 4	Next sequential PC from the Fetch stage.
IF_instruction	32-bit Input	Instruction Memory	Raw instruction fetched from memory.
IF_branch_estimation	1-bit Input	Branch Predictor	Prediction bit from the Fetch stage.

Port Name	**Width / Type**	**Destination Module**	**Explanation**
ID_pc	32-bit Output	ID/EX Register	PC value passed to the Decode stage.
ID_pc_plus_4	32-bit Output	ID/EX Register	xt sequential PC passed to the Decode stag
ID_instruction	32-bit Output	Instruction Decoder	Instruction passed to the Decode stage.

ID_branch_estimation	1-bit Output	ID/EX Register	Prediction bit passed to the Decode stage.

Instruction Decoder (InstructionDecoder.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
instruction	32 bits	IF/ID Register	Fetched machine code instruction to be decoded

Output Ports:

Port Name	Width	Destination Module	Explanation
opcode	7 bits	Control Unit / ID Logic	Major operation code extracted from instruction[6:0]
funct3	3 bits	Control Unit / ALU Contro	Function field 3 extracted from instruction[14:12]
funct7	7 bits	Control Unit / ALU Contro	Function field 7 extracted from instruction[31:25]
rs1	5 bits	Register File	ource register 1 index extracted from instruction[19:15
rs2	5 bits	Register File	ource register 2 index extracted from instruction[24:20
rd	5 bits	ID/EX Register	estination register index extracted from instruction[11:

raw_imm	20 bits	Immediate Generator	ıcted unextended immediate bits (scrambled based on '

Immediate Generator (ImmediateGenerator.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
raw_imm	20 bits	Instruction Decoder	Raw scrambled immediate bits from the decoder
opcode	7 bits	Instruction Decoder	Opcode used to determine the sign-extension format

Output Ports:

Port Name	Width	Destination Module	Explanation
imm	32 bits	ID/EX Register	Fully formatted, sign-extended 32-bit immediate value

ID/EX Pipeline Register (ID_EX_Register.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
clk	1 bit	Top Module	System clock
reset	1 bit	Top Module	Synchronous reset

flush	1 bit	Hazard Unit	nes register content (inserts N
ID_EX_stall	1 bit	Hazard Unit	the pipeline (holds previous
ID_pc	32 bits	IF/ID Register	rogram Counter from ID stag
ID_pc_plus_4	32 bits	ID Adder	ext PC (PC + 4) from ID stag
ID_branch_estimation	1 bit	Branch Predictor	nch prediction bit from ID st
ID_instruction	32 bits	IF/ID Register	instruction word being proce
ID_jump	1 bit	Control Unit	Jump control signal
ID_branch	1 bit	Control Unit	Branch control signal
ID_alu_src_A_select	2 bits	Control Unit	lect signal for ALU operand
ID_alu_src_B_select	3 bits	Control Unit	elect signal for ALU operand
ID_memory_read	1 bit	Control Unit	Memory read enable signal
ID_memory_write	1 bit	Control Unit	Memory write enable signal

ID_register_file_write_data_select	3 bits	Control Unit	elect signal for Write Back da
ID_register_write_enable	1 bit	Control Unit	egister file write enable sign
ID_csr_write_enable	1 bit	Control Unit	CSR write enable signal
ID_opcode	7 bits	Instruction Decoder	Opcode passed from ID stage
ID_funct3	3 bits	Instruction Decoder	nct3 field passed from ID sta
ID_funct7	7 bits	Instruction Decoder	nct7 field passed from ID sta
ID_rd	5 bits	Instruction Decoder	Destination register address
ID_raw_imm	20 bits	Instruction Decoder	Raw immediate bits
ID_read_data1	32 bits	Register File	ata read from source register
ID_read_data2	32 bits	Register File	ata read from source register
ID_rs1	5 bits	Instruction Decoder	Source register 1 address
ID_rs2	5 bits	Instruction Decoder	Source register 2 address

ID_imm	32 bits	mmediate Generato	ign-extended immediate valu
ID_csr_read_data	32 bits	CSR Unit	Data read from CSR register

Output Ports:

Port Name	Width	Destination Module	Explanation
EX_pc	32 bits	EX Stage	PC passed to Execute stage
EX_pc_plus_4	32 bits	EX Stage	xt PC passed to Execute st
EX_branch_estimation	1 bit	EX Branch Logic	anch estimation passed to I
EX_instruction	32 bits	EX Stage	struction word passed to E
EX_jump	1 bit	EX Stage	Jump signal passed to EX
EX_memory_read	1 bit	EX/MEM Register	mory read signal passed to
EX_memory_write	1 bit	EX/MEM Register	mory write signal passed to
EX_register_file_write_data_select	3 bits	EX/MEM Register	selection signal passed to

EX_register_write_enable	1 bit	EX/MEM Register	eg write enable passed to E
EX_csr_write_enable	1 bit	EX/MEM Register	R write enable passed to E
EX_branch	1 bit	EX Branch Logic	Branch signal passed to EX
EX_alu_src_A_select	2 bits	ALU Mux	U A select signal passed to
EX_alu_src_B_select	3 bits	ALU Mux	U B select signal passed to
EX_opcode	7 bits	ALU Control	Opcode passed to EX
EX_funct3	3 bits	ALU Control	Funct3 passed to EX
EX_funct7	7 bits	ALU Control	Funct7 passed to EX
EX_rd	5 bits	rwarding Unit / EX/ME	st register addr passed to E
EX_raw_imm	20 bits	EX Stage	aw immediate passed to E
EX_read_data1	32 bits	ALU / Forwarding Unit	RS1 data passed to EX
EX_read_data2	32 bits	ALU / Forwarding Unit	RS2 data passed to EX

EX_rs1	5 bits	Forwarding Unit	RS1 address passed to EX
EX_rs2	5 bits	Forwarding Unit	RS2 address passed to EX
EX_imm	32 bits	ALU	ended immediate passed to
EX_csr_read_data	32 bits	EX Stage	CSR data passed to EX

HazardUnit (HazardUnit.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
clk	1 bit	Top Module	ystem clock used for updating internal registers
reset	1 bit	Reset Controller	Resets internal tracking registers.
trap_done	1 bit	Trap_Controller	Indicates trap handling is finished.
csr_ready	1 bit	CSR_File	Shows CSR operations are complete.
standby_mode	1 bit	Exception_Detector	Pauses pipeline during exception handling.
trap_status	3 bits	Trap_Controller	Provides current trap state information.
ligned_instruction_f	1 bit	Exception_Detector	Indicates instruction alignment error.
saligned_memory_flu	1 bit	Exception_Detector	Indicates memory alignment error.
pth_done_flush	1 bit	Trap_Controller	Indicates pre-trap handling completion.
ID_rs1	5 bits	IF_ID_Pipeline_Reg	Source register 1 from Decode stage.
ID_rs2	5 bits	IF_ID_Pipeline_Reg	Source register 2 from Decode stage.
ID_raw_imm	12 bits	Instruction_Decoder	Immediate value from instruction.
EX_rd	5 bits	ID_EX_Pipeline_Reg	Destination register in EX stage.
EX_opcode	7 bits	ID_EX_Pipeline_Reg	Opcode of instruction in EX stage.
EX_rs1	5 bits	ID_EX_Pipeline_Reg	Source register 1 in EX stage.

EX_rs2	5 bits	ID_EX_Pipeline_Reg	Source register 2 in EX stage.
EX_imm	12 bits	Immediate_Generator	Immediate value used in EX stage.
EX_csr_write_enable	1 bit	Control_Unit	Indicates EX stage writes to CSR.
EX_jump	1 bit	Branch_Logic	Indicates jump instruction execution.
MEM_rd	5 bits	EX_MEM_Pipeline_Reg	Destination register in MEM stage.
MEM_reg_write_en	1 bit	Control_Unit	Indicates MEM stage writes to register file.
MEM_csr_write_en	1 bit	Control_Unit	Indicates MEM stage writes to CSR.
MEM_csr_addr	12 bits	EX_MEM_Pipeline_Reg	CSR address written in MEM stage.
WB_rd	5 bits	MEM_WB_Pipeline_Reg	Destination register in WB stage.
WB_reg_write_en	1 bit	Control_Unit	Indicates WB stage writes to register file.
WB_csr_write_en	1 bit	Control_Unit	Indicates WB stage writes to CSR.
WB_csr_addr	12 bits	MEM_WB_Pipeline_Reg	CSR address written in WB stage.
branch_pred_miss	1 bit	Branch_Predictor	Indicates incorrect branch prediction.

Output Ports:

Port Name	Width	Destination Module	Explanation
hazard_mem	2 bits	ForwardUnit	ALU operand forwarding from MEM stage.
hazard_wb	2 bits	ForwardUnit	ALU operand forwarding from WB stage.
csr_hazard_mem	1 bit	CSR_File	CSR dependency with MEM stage.
csr_hazard_wb	1 bit	CSR_File	CSR dependency with WB stage.
store_hazard_mem	1 bit	ForwardUnit	Store data forwarding from MEM stage.
store_hazard_wb	1 bit	ForwardUnit	Store data forwarding from WB stage.
IF_ID_flush	1 bit	IF_ID_Pipeline_Reg	Clears instruction in IF/ID stage.
ID_EX_flush	1 bit	ID_EX_Pipeline_Reg	Clears instruction in ID/EX stage.
EX_MEM_flush	1 bit	EX_MEM_Pipeline_Reg	Clears instruction in EX/MEM stage.
MEM_WB_flush	1 bit	MEM_WB_Pipeline_Reg	Clears instruction in MEM/WB stage.
IF_ID_stall	1 bit	IF_ID_Pipeline_Reg	Stops instruction fetch/decode.
ID_EX_stall	1 bit	ID_EX_Pipeline_Reg	Pauses execution at EX entry.
EX_MEM_stall	1 bit	EX_MEM_Pipeline_Reg	Pauses MEM stage pipeline flow.
MEM_WB_stall	1 bit	MEM_WB_Pipeline_Reg	Pauses WB stage pipeline flow.

ForwardUnit (Forward_Unit.v)

Input Ports:

Port Name	Width	Source Module	Explanation
hazard_mem	2 bits	HazardUnit	Forwarding requirement from MEM stage.
hazard_wb	2 bits	HazardUnit	Forwarding requirement from WB stage.
store_hazard_mem	1 bit	HazardUnit	Store data forwarding required from MEM.
store_hazard_wb	1 bit	HazardUnit	Store data forwarding required from WB.
MEM_imm	32 bits	EX_MEM_Pipeline_Reg	Immediate value (e.g., LUI) in MEM stage.
MEM_alu_result	32 bits	EX_MEM_Pipeline_Reg	ALU result produced in MEM stage.
MEM_csr_read_data	32 bits	CSR_File	CSR read value in MEM stage.
te_enable_logic_rf_v	32 bits	Byte_Enable_Logic	Load data after byte selection/extension.
MEM_pc_plus_4	32 bits	EX_MEM_Pipeline_Reg	PC+4 value in MEM stage.
MEM_opcode	7 bits	EX_MEM_Pipeline_Reg	Opcode to identify instruction in MEM.
WB_imm	32 bits	MEM_WB_Pipeline_Re	Immediate value (e.g., LUI) in WB stage.
WB_alu_result	32 bits	MEM_WB_Pipeline_Re	ALU result in WB stage.
WB_csr_read_data	32 bits	CSR_File	CSR read value in WB stage.
WB_be_logic_rf_wd	32 bits	Byte_Enable_Logic	Load data after byte selection in WB stage.
WB_pc_plus_4	32 bits	MEM_WB_Pipeline_Re	PC+4 value in WB stage.
WB_opcode	7 bits	MEM_WB_Pipeline_Re	Opcode to identify instruction in WB.
csr_hazard_mem	1 bit	HazardUnit	CSR hazard caused by MEM stage.
csr_hazard_wb	1 bit	HazardUnit	CSR hazard caused by WB stage.
MEM_csr_write_data	32 bits	CSR_File	CSR data being written in MEM stage.
WB_csr_write_data	32 bits	CSR_File	CSR data being written in WB stage.
csr_read_data	32 bits	CSR_File	Normal CSR read output (no hazard).

Output Ports:

Port Name	Width	Destination Module	Explanation
_forward_source_dat	32 bits	ALU_Source_MUX_A	Forwarded operand data for ALU input A.
_forward_source_dat	32 bits	ALU_Source_MUX_B	Forwarded operand data for ALU input B.
forward_source_sele	2 bits	ALU_Source_MUX_A	Select signal for ALU input A source.ID_EX
forward_source_sele	2 bits	ALU_Source_MUX_B	Select signal for ALU input B source.
store_forward_data	32 bits	Data_Memory	Forwarded data for store instructions.
tore_forward_enabl	1 bit	Data_Memory	Enables forwarding of store data.
csr_forward_data	32 bits	CSR_File	Forwarded CSR data during hazards.

Trap Controller Unit (Trap_Controller.v)

Input Ports:

Port Name	Width	Source Module	Explanation
clk	1 bit	Top Module	System clock for the TrapController FSM.
reset	1 bit	Reset Controller	Global reset for FSM and internal registers.
ID_pc	32 bits	IF_ID_Pipeline_Reg	PC value of instruction in Decode stage.
EX_pc	32 bits	ID_EX_Pipeline_Reg	PC value of instruction in Execute stage.
MEM_pc	32 bits	EX_MEM_Pipeline_Reg	PC value of instruction in Memory stage.
WB_pc	32 bits	MEM_WB_Pipeline_Re	PC value of instruction in Writeback stage.
trap_status	3 bits	Exception_Detector	Encoded trap type (misaligned, ecall, mret).
csr_read_data	32 bits	CSR_File	Data read from CSR during trap handling.

Output Ports:

Port Name	Width	Destination Module	Explanation
trap_target	32 bits	PC_Controller	Base address for trap handler or return target.
ic_clean	1 bit	Instruction_Cache	Invalidate/clean instruction cache (FENCE.I).
debug_mode	1 bit	Debug_Module	ndicates entry into debug mode (e.g., EBREAK
csr_write_enable	1 bit	CSR_File	Requests a CSR write operation.
csr_trap_address	12 bits	CSR_File	CSR address for read/write (mtvec, mepc, etc.).
csr_trap_write_data	32 bits	CSR_File	Data to write into the selected CSR.
trap_done	1 bit	HazardUnit	Indicates Pre-Trap Handling (PTH) is finished.
misaligned_inst_flush	1 bit	HazardUnit	Flush pipeline for misaligned-instruction trap.
misaligned_mem_flus	1 bit	HazardUnit	Flush pipeline for misaligned memory access.
pth_done_flush	1 bit	HazardUnit	ide-flush once PTH completes and target is rea
standby_mode	1 bit	HazardUnit	all pipeline during multi-cycle ECALL handlin

Branch Predictor (Branch_Predictor.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
clk	1 bit	Top Module	Clock signal used to update the prediction state

reset	1 bit	Top Module	Resets branch prediction history
IF_opcode	7 bits	Instruction Fetch stage	Opcode of the instruction in IF stage
IF_pc	XLEN	Program Counter	Current program counter value
IF_imm	XLEN	ediate Generator (IF st	Immediate offset for branch target calculation
EX_branch	1 bit	Execute stage	Indicates a branch instruction in EX stage
EX_branch_taken	1 bit	Branch Logic	Actual result of the branch execution

Output Ports:

Output Signal	Width	Destination Module	Description
branch_estimation	1	nstruction Fetch (IF) stag	Predictor’s decision: predict branch taken or no
branch_target	XLEN	nstruction Fetch (IF) stag	Predicted next PC value

CSR File (CSR_File.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
clk	1 bit	Top Module	Clock signal for CSR updates
reset	1 bit	Top Module	Resets all CSR registers
trapped	1 bit	Trap / Exception Logic	Indicates an exception or trap occurred
csr_write_enable	1 bit	ntrol Unit / Trap Control	Enables writing to CSR registers
csr_read_address	12 bits	Decode / WB stage	Address of CSR to be read
csr_write_address	12 bits	WB / Trap Controller	Address of CSR to be written
csr_write_data	XLEN	ALU / Trap Controller	Data to write into CSR
instruction_retired	1 bit	Writeback stage	Indicates instruction completion

Output Ports:

Port Name	Width	Destination Module	Explanation
csr_read_out	XLEN	ALU / Register File	Data read from CSR register
csr_ready	1 bit	Hazard / Control Unit	Indicates CSR access completion

Exception Detector (Exception_Detector.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
clk	1 bit	Top Module	Clock for synchronous trap output
reset	1 bit	Top Module	Resets trap signals
ID_opcode	7 bits	ID Stage	Opcode in Instruction Decode stage
EX_opcode	7 bits	EX Stage	Opcode in Execute stage
MEM_opcode	7 bits	MEM Stage	Opcode in Memory stage
ID_funct3	3 bits	ID Stage	funct3 field from ID stage
EX_funct3	3 bits	EX Stage	funct3 field from EX stage
MEM_funct3	3 bits	MEM Stage	funct3 field from MEM stage
alu_result	2 bits	ALU (EX stage)	LSBs of calculated address
MEM_alu_result	2 bits	ALU (MEM stage)	LSBs of memory address
raw_imm	12 bits	ID Stage	Immediate for ECALL / EBREAK / CSR
EX_raw_imm	12 bits	EX Stage	Immediate field in EX stage
csr_write_enable	1 bit	Control Unit	Used for illegal CSR detection
branch_target_lsbs	2 bits	Branch Logic	LSBs of branch target address
branch_estimation	1 bit	Branch Predictor	Indicates predicted branch taken

Output Ports:

Port Name	Width	Destination Module	Explanation
trapped	1 bit	Trap Controller / Hazard Unit	Indicates an exception or trap occurred
trap_status	3 bits	Trap Controller	Encodes the reason for the trap

Arithmetic Logic Unit (ALU.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
src_A	32 bits	ID_EX_Register	Source operand A
src_B	32 bits	ID_EX_Register	Source operand B
alu_op	4 bits	ALU_Controller	ol signal to select the instru

Output Ports:

Port Name	Width / Type	Destination Module	Explanation
alu_result	32 bits	RV32I46F_5SP	Result of operation
alu_zero	1 bit	anch Logic, RV32I46F_5S	Zero flag if alu_result is 0

ALU Controller (ALU_Controller.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
opcode	7 bits	ID_EX_Register	Instruction opcode
funct3	3 bits	ID_EX_Register	between variants of the sa instruction
funct7_5	1 bit	ID_EX_Register	h between ADD/SUB and R-type instructions
imm_10	1 bit	ID_EX_Register	guish between SRLI/SRAI instructions

Output Ports:

Port Name	Width / Type	Destination Module	Explanation
alu_op	5 bits	ALU	ol signal to select the instru

Branch Logic (Branch_Logic.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
branch	1 bit	ID_EX_Register	ndicates a branch instructio
branch_estimation	1 bit	ID_EX_Register	Prediction from Fetch stage
alu_zero	1 bit	ALU	Zero flag from ALU
funct3	3 bits	ID_EX_Register	Branch condition type
pc	32 bits	ID_EX_Register	Program Counter
imm	32 bits	ID_EX_Register	Branch offset (immediate)

Output Ports:

Port Name	Width / Type	Destination Module	Explanation
branch_taken	1 bit	RV32I46F_5SP	ue if branch condition is m
branch_target_actual	32 bits	RV32I46F_5SP	Calculated target address
branch_prediction_miss	1 bit	RV32I46F_5SP	rediction misprediction fla

Execute / Memory Access Pipeline Register (EX_MEM_Register.v)

Input Ports:

Port Name	Width / Type	Source Module	Explanation
clk	1 bit	Top Module	System clock
reset	1 bit	Top Module	System reset
flush	1 bit	Hazard Unit	Pipeline flush signal
EX_MEM_stall	1 bit	Hazard Unit	Pipeline stall signal
EX_pc	32 bits	ID_EX_Register	PC from EX stage
EX_pc_plus_4	32 bits	ID_EX_Register	Next PC from EX stage
EX_instruction	32 bits	ID_EX_Register	Instruction word
EX_memory_read	1 bit	ID_EX_Register	Memory read enable
EX_memory_write	1 bit	ID_EX_Register	Memory write enable
egister_file_write_data_s	3 bits	ID_EX_Register	Write back data mux select
X_register_write_enable	1 bit	ID_EX_Register	Register file write enable
EX_csr_write_enable	1 bit	ID_EX_Register	CSR write enable
EX_opcode	7 bits	ID_EX_Register	Opcode
EX_funct3	3 bits	ID_EX_Register	Funct3
EX_rs1	5 bits	ID_EX_Register	Source register 1 address
EX_rd	5 bits	ID_EX_Register	Destination register address
EX_read_data2	32 bits	ID_EX_Register	Store data (rs2 value)
EX_imm	32 bits	ID_EX_Register	Immediate value
EX_raw_imm	20 bits	ID_EX_Register	Raw immediate
EX_csr_read_data	32 bits	ID_EX_Register	CSR read data
EX_alu_result	32 bits	EX Phase	LU result / Memory Addre

Output Ports:

Port Name	Width / Type	Destination Module	Explanation
MEM_pc	32 bits	EX_MEM_Register	PC to MEM stage
MEM_pc_plus_4	32 bits	EX_MEM_Register	Next PC to MEM stage
MEM_instruction	32 bits	EX_MEM_Register	Instruction to MEM stage
MEM_memory_read	1 bit	EX_MEM_Register	Mem read enable to MEM
MEM_memory_write	1 bit	EX_MEM_Register	Mem write enable to MEM
_register_file_write_data_	3 bits	EX_MEM_Register	WB Mux sel to MEM
EM_register_write_enab	1 bit	EX_MEM_Register	Reg write enable to MEM
MEM_csr_write_enable	1 bit	EX_MEM_Register	CSR write enable to MEM
MEM_opcode	7 bits	EX_MEM_Register	Opcode to MEM
MEM_funct3	3 bits	EX_MEM_Register	Funct3 to MEM
MEM_rs1	5 bits	EX_MEM_Register	RS1 addr to MEM
MEM_rd	5 bits	EX_MEM_Register	Dest reg addr to MEM
MEM_read_data2	32 bits	EX_MEM_Register	Store data to MEM
MEM_imm	32 bits	EX_MEM_Register	Imm to MEM
MEM_raw_imm	20 bits	EX_MEM_Register	Raw imm to MEM
MEM_csr_read_data	32 bits	EX_MEM_Register	CSR data to MEM

Module: Byte_Enable_Logic

Input Ports

Port Name	Width / Type	Source Module	Explanation
mem_read_data	32 bits	Data Memory	Raw 32-bit memory output
alu_result	32 bits	EX Stage	Memory address used for byte offset selection
funct3	3 bits	ID Stage	Determines load type (LB, LBU, LH, LHU, LW)

Output Ports:

Port Name	Width	Destination Module	Explanation
processed_load_data	32 bits	MEM/WB Register	perly formatted and extended load data for writeb

Module : Data_Memory

Input Ports:

Port Name	Width / Type	Source Module	Explanation
address	32 bits	ALU (EX Stage)	Memory address for load/store
write_data	32 bits	Register File (via pipeline)	Data to be stored in memory
write_enable	1 bit	Control Unit	Enables store operation
byte_enable	4 bits	Control Logic	Controls partial byte writes (SB, SH, SW)

Output Ports:

Port Name	Width	Destination Module	Explanation
read_data	32 bits	ByteEnableLogic	Raw 32-bit memory read data
exception_signal	1 bit	Exception Detector	Indicates misaligned or illegal access

MODULE: Instruction_Memory

Input Ports

Port Name	Width / Type	Source Module	Explanation
PC	32 bits	Program Counter	Instruction address input
rom_debug_addr	32 bits	Debug Interface	Direct ROM access for debugging

Output Ports:

Port Name	Width	Destination Module	Explanation
instruction	32 bits	IF/ID Register	Instruction to decode
rom_read_data	32 bits	Debug Interface	External ROM read access

MODULE: MEM_WB_Register

Input Ports

Port Name	Width / Type	Source Module	Explanation
mem_pc	32 bits	EX/MEM Register	Program counter value
mem_instruction	32 bits	EX/MEM Register	Current instruction
mem_alu_result	32 bits	ALU	ALU computation result
mem_load_data	32 bits	ByteEnableLogic	Formatted load data
mem_reg_write	1 bit	Control Unit	Register write enable
mem_rd	5 bits	Decode Stage	Destination register index

clk	1 bit	Top Module	Clock signal
reset	1 bit	Top Module	Pipeline reset
stall	1 bit	Hazard Unit	Pipeline stall control
flush	1 bit	Control Logic	Pipeline flush control

Output Ports:

Port Name	**Width**	**Destination Module**	**Explanation**
wb_pc	32 bits	WB Stage	PC value for writeback stage
wb_instruction	32 bits	WB Stage	Instruction in WB stage
wb_alu_result	32 bits	WB MUX	ALU result forwarded
wb_load_data	32 bits	WB MUX	Load data forwarded
wb_reg_write	1 bit	Register File	Write enable signal
wb_rd	5 bits	Register File	Destination register index

ControlUnit

Input ports:

Port Name	**Width / Type**	**Source Module**	**Explanation**
write_done	1-bit / Input	External / Datapath	flag indicating if a pending write oper completed.

trap_done	1-bit / Input	TrapController	s flag indicating that Pre-Trap Handli successfully finished.
csr_ready	1-bit / Input	CSRFile	; indicating the Control and Status Re ready for operations.
IF_ID_stall	1-bit / Input	HazardUnit	ol signal indicating that the Instruction ecode pipeline stage is currently stalle
opcode	7-bit / Input	InstructionDecoder	mary 7-bit operational code from the tion, used to determine the instruction
funct3	3-bit / Input	InstructionDecoder	ry 3-bit function code from the instruc ecoding (e.g., distinguishing specific CSR instructions).

Output **ports:**

		Destination Module	**Explanation**
jump	1-bit / Output	PCController / Datapath	when an unconditional jump instruction (J is decoded.

branch	1-bit / Output	BranchLogic / Datapath	gh when a conditional branch instruction (B etc.) is decoded.
alu_src_A_select	2-bit / Output	LU Input A Multiplexe	the data source for ALU input A (e.g., Reg Program Counter, or zero).
alu_src_B_select	3-bit / Output	LU Input B Multiplexe	the data source for ALU input B (e.g., Reg Immediate, Shift Amount, or CSR data).
csr_write_enable	1-bit / Output	CSRFile / Datapath	riting new data into the Control and Status
register_file_write	1-bit / Output	RegisterFile	riting data back into the destination register main Register File.

er_file_write_data_	3-bit / Output	Write-Back Multiplexer	e source of the data to be written to the Reg ALU result, Memory Load, Immediate, or F
memory_read	1-bit / Output	Memory / ByteEnableL	high to instruct the Data Memory to perforr operation (used for LOAD instructions).
memory_write	1-bit / Output	Memory / ByteEnableL	high to instruct the Data Memory to perforr operation (used for STORE instructions).
pc_stall	1-bit / Output	PCController	igh to freeze the Program Counter when ha unready or a pipeline stall is active.